

PROJECT TRICORDER

MK I · FIELD OPERATIONS MANUAL · REV 0.6.21

On-device multimodal field instrument. Inquiry, vision, image synthesis, neural speech in/out, offline translator, audio scene analysis, flora/fauna, sentry watch, and spoken dictation with a token-buffer editor.

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1. Purpose and scope

This is the operator and installer document for Project Tricorder Mk I through software 0.6.21. It supersedes Rev 0.6.18 (9 Sep 2026). Material from 0.6.4–0.6.14 remains in force unless a later row in §2 replaces it. The on-device Help sheet is a short briefing. This file is the full operations manual.

NAVIGATION WARNING. The artificial horizon, G-meter, GPS speed, satellite plot, star map, SOL orbits, and HF/VHF/UHF forecasts are NOT certified by FAA, NASA, EASA, or ICAO. They MUST NEVER be used as primary navigation for any aircraft, spacecraft, or surface vessel.

SENTRY / AUDIO WATCH WARNING. Weapon, gunfire, and conflict alerts are keyword hits on SmolVLM captions or YAMNet AudioSet labels. They are not a weapons detector, not a 911 substitute, and will false-positive on toys, pictures, fireworks, and media.

DICTATION NOTE. Dictation uses the same on-device Android SpeechRecognizer as Inquiry. Titles, body text, and command words are only as good as that recognizer. Speak command verbs in isolation when possible.

2. What changed since Rev 0.6.14

Rev 0.6.14 closed the audio soundscape watch. Builds 0.6.15–0.6.17 add a buttonless spoken memo system that does not require a new Termux backend.

Build	Change
0.6.15	Wake word DICTATION. Directory of numbered memos. NEW / EDIT / SUMMARIZE / MINDMAP / SPEAK / KILL / TERMINATE. ZULU menu. BACKSPACE and NEWLINE. Local JSON store + synset embeddings. USE embedder fetched on first Earth-link boot.
0.6.16	Wake word actually intercepted before Gemma. 0.6.15 shipped the engine but Inquiry swallowed the word “dictation” and answered it as chat. Isolated utterances: dictation, dictation mode, start/enter dictation.
0.6.17	Memo held as a JSON token array painted on the CRT with a visible  cursor. CURSOR MINUS/PLUS keyword. CURSOR CUT / CUT WORD / PASTE. CURSOR HOME / END. BACKSPACE deletes the token before the cursor.
0.6.18	LOG nnnn in the dictation directory. Prompt: convert nnnn %title% to tricorder log entry? CONFIRMED appends kind dictation to the field log (title + body + GPS/UTC). ABORT cancels.

	Memo stays in the dictation store either way. CRT and LED menus list LOG with NEW/EDIT.
0.6.19	Dictation EDIT: PUNCTUATION period/dot/comma/semicolon/question/colon/exclamation attaches the mark to the word before the cursor. Bare period, dot, comma, semicolon, question also convert.
0.6.20	Native IMU feed so the artificial horizon moves at boot. BARO hPa and pressure-altitude proxy plus GPS altitude on the tape. Aircraft alerts use robot voice and convert m / m/s to meters+km/h or feet+mph. IPL sliders labeled ALL and STT. ROBO toggle forces cheap TTS for all Inquiry speech. Say DIAGNOSTICS in Inquiry for version, packs, sensors, credits, and trademark disclaimer.
0.6.21	Flight watch looks up departure and destination (adsbdb callsign, OpenSky flights/aircraft fallback) and speaks from CITY IATA to CITY IATA after the altitude/speed clause.

Unchanged from 0.6.14: sentry weapon tiers, continuous III, IPL + Kokoro-kill sliders, 80-character Kokoro chunks, Sensor/Sentry dwell sliders, first-frame capture, audio soundscape watch, translator, horizon, Sat/Stars, SOL cycle, field log, mesh alerts.

3. Face controls (0.6.17)

Control	Function
I Inquiry	Gemma chat. Imagine keywords. COMPUTER easter egg. Exact HELP opens the spoken briefing. Isolated DICTATION enters dictation mode.
II Sensor	Camera + SmolVLM + flora/fauna. Weapon watch on the caption. First frame as soon as the camera is ready; dwell is the gap between frames.
III Sentry	Continuous by default. Captions every dwell. Sure weapon: klaxon + WEAPON DETECTED x3. Motion is logged only if the caption has no watch hit.
IV Audio	YAMNet duty cycle, BirdNET when the clip looks avian, context watches if checked. Soundscape watch (\$6 of Rev 0.6.14).
T Translator	On-device ML Kit speech-to-speech. SOURCE / TARGET wheels. AUTO swaps direction after each line.
H Horizon	Artificial horizon, heading, G-meter, GPS speed. Not for flight.
S Sat/Stars	Celestrak TLE / SGP4 satellites, then major stars. Live from phone pose and time.
O Sol	Press 1: solar-system + space weather + HF/VHF/UHF. Press 2: local earth forecast. Press 3: off.
IPL slider	Aborts in-flight chat / vision / Imagine / voice / translate. Lamp green while resetting, red after trip.

Kokoro slider	Kills only in-flight Kokoro. Does not abort image gen or vision.
L E V P	Log review slider, officer audio log, video log, playback.

4. Dictation mode — theory of operations (0.6.15–0.6.21)

4.1 Why it exists

Field notes on a phone that already has a mic, a CRT, and a local language model should not require a separate TikTok-shop note-taker. The operator keeps both hands on the instrument. There is no new dictation button. The wake word is the button.

4.2 What was not reused

Open-source cousins (Scrib, Dicta, Whisper IME, Vox Transcribe, OpenWispr) are keyboards or record-then-summarize apps. None implement a wake-word directory, ZULU commit, keyword cursor, and semantic mindmap on a field face. Tricorder therefore owns the state machine in the APK. Speech-to-text is Android’s on-device SpeechRecognizer (same path as Inquiry). Storage is JSON under the app files directory. Summarize will use Gemma on 127.0.0.1:8080 when chat is up; otherwise it stores the first sentences locally. No fifth Termux port.

4.3 Entering and leaving

Arm Inquiry (I). Press the mic. Say only one of:

dictation · dictation mode · start dictation · enter dictation

The robot voice confirms. The CRT lists memos as nnnn title. The LED matrix repeats legal verbs every ten seconds. Say TERMINATE to return to Inquiry. A longer sentence that merely contains the word “dictation” is still sent to Gemma.

4.4 Directory verbs

Say	Result
NEW	Robot asks for a title. The next utterance becomes the title. Memo id is the next four-digit number. Edit opens.
EDIT nnnn	Load that memo. Spoken numbers work (edit seven → 0007).
SUMMARIZE nnnn	Creates a new memo titled “summarized: %title%”. Gemma if :8080 answers; else extractive local summary.
MINDMAP keyword	Related memos by synset + 128-d hashed vectors. puzzle also hits riddle and crossword. Touch a node to edit.
SPEAK nnnn	Kokoro reads title and body (chunked over 80 characters).
KILL nnnn	Requires CONFIRMED. Anything else leaves the memo in place.
TERMINATE	Leave dictation. Inquiry mode.
LOG nnnn	Prompt: convert nnnn %title% to tricorder log entry? Say CONFIRMED or ABORT. CONFIRMED

	appends kind dictation to the tricorder field log (id, title, body, timestamp, GPS if fixed). Review on the log slider. The memo is not deleted.
PUNCTUATION period (or comma, semicolon, question)	In EDIT, attaches . , ; ? : ! to the token before the cursor. Bare PERIOD / DOT / COMMA / SEMICOLON / QUESTION also work.
DIAGNOSTICS	Inquiry intercept. Robot voice reads software version, device, battery, earth link, baro, compass, accel/gyro, backend lamps, BirdNET/flora/translator/Termux state, recent init fetches, capabilities, models, N2NHU credit, and the unaffiliated-trademark disclaimer.

4.5 Edit / ZULU

While editing, speech is body text unless it is a reserved verb. Say ZULU to leave speak-to-text and open the menu. ZULU then accepts SAVE (writes the joined buffer to disk and returns to the directory) or ABORT (requires CONFIRMED; edits are discarded).

BACKSPACE deletes the token immediately before the cursor — not the word “backspace”. If you just landed on ROVER with CURSOR MINUS, BACKSPACE removes ROVER. NEWLINE / new line inserts a line break at the cursor.

4.6 Token buffer and CRT (0.6.17)

The live document is a JSON array of tokens (words and explicit newline markers) plus an integer cursor. The CRT is a projection of that array. A block glyph █ marks the insertion point. The header shows @index/length and, when a cut is pending, a CLIP preview. SAVE serializes the array back to a single body string in memos.json. The cursor is session state; it is not persisted across TERMINATE.

4.7 Cursor verbs (0.6.17)

Say	Cursor / buffer
CURSOR MINUS rover	Search backward from the cursor for rover (case-insensitive). Rest after that token.
CURSOR PLUS rover	Search forward. Same rest point. Aliases: back / previous / forward / next.
CURSOR CUT	Cut from the cursor through the next sentence (period, question, bang, or newline).
CURSOR CUT WORD	Cut only the token after the cursor.
CURSOR PASTE	Insert the clip at the cursor and advance past it. One clip; next CUT replaces it.
CURSOR HOME	Cursor to index 0.
CURSOR END	Cursor to the end of the buffer.
BACKSPACE	Delete the token before the cursor.
spoken words	Insert at the cursor.

Example. Body: the red rover rolled past the rover bay. Say CURSOR MINUS rover — cursor sits after the second rover. BACKSPACE removes that rover. Dictate “scout” — scout appears in its place. CURSOR CUT takes the rest of that sentence into CLIP. Move with CURSOR END and CURSOR PASTE to drop it at the end.

4.8 Mindmap semantics

Each memo stores a 128-dimension feature-hash of title + body after synonym expansion. Built-in synsets bind puzzle/riddle/crossword, weapon/gun/pistol, weather/storm/rain, plant/flora/tree, and similar field clusters. Cosine similarity plus a literal-token bonus ranks hits. On first Earth-link boot the APK also fetches MediaPipe’s Universal Sentence Encoder into files/models so a later build can swap the hasher for a true sentence embedding without another Termux service. Matching already works offline if that file is absent.

4.9 Limits

The recognizer may hear “curse or minus” or split BACKSPACE as two tokens; the editor accepts “back space”. Keyword search is exact after punctuation strip — rover matches Rover, not roving. Summaries depend on Gemma being up. Memos live in the app private directory; uninstall wipes them. Dictation pauses the usual Inquiry path for that session only.

5. Sentry and audio watch (retained from 0.6.14)

III on a rock at a cave mouth remains the intended sentry use. Continuous vision writes a caption every dwell. Sure-weapon tokens (pistol, handgun, revolver, rifle, shotgun, firearm, Glock, AK-47, AR-15, knife, dagger, machete, sword) raise klaxon + WEAPON DETECTED ×3 on the cheap robot voice, log kind sentry_weapon. Dual-use (bat, pipe, chain, club) and violence verbs raise a chime. Motion is a changed-pixel fraction (~7%) and never gates the caption; motion is spoken only when the caption has no watch hit (20 s cool). II Sensor runs the same weapon watch on its caption.

IV Audio maps YAMNet labels through a priority table: gunfire ×3 + klaxon; scream/shout; siren ×3; air / rail / ship; road vehicles; named animal; humans ×3; entertainment device. Hits are logged. Speech and media have long cooldowns so the operator’s own voice and Kokoro do not retrigger. Fireworks can look like gunfire. Not a 911 path.

6. IPL, Kokoro, Imagine, dwell (retained)

IPL slider aborts in-flight chat, vision, Imagine, voice, and translate. Lamp green while resetting, red after the trip. The Kokoro slider kills only TTS. Text longer than 80 characters is split on sentence then word boundaries before Kokoro. Settings still expose separate Sensor and Sentry dwell (default 3 s). The first camera frame is sent as soon as the preview is ready; a long first look is SmolVLM inference, not a hidden 10 s timer.

Imagine sliders remain step count and size (256 / 384 / 512) against the turbo checkpoint at cfg-scale 1. Four steps at 384 or 512 is the quality/speed region that survived field tests on Snapdragon 888.

7. Field log kinds

kind	When
officer / video / sensor / sentry	Manual or mode captures (unchanged).

sentry_weapon / sentry_possible / sentry_scene	Sentry watch tiers.
audio_*	Soundscape classes from IV.
dictation memos	Separate store: filesDir/dictation/memos.json. Not mixed into the field-log slider.

8. Architecture recap

Kotlin WebView APK talks to four Termux processes on loopback: Gemma chat :8080, SmolVLM :8081, sd-turbo :8188, Kokoro :5000. On-device TFLite covers YAMNet, BirdNET, iNaturalist flora/fauna. ML Kit covers translation language packs. Nearby Connections carries camp-to-hole alerts. Dictation, horizon, SOL, TLE, and the translator UI live in the APK. Do not stop 8188 to free a port — all four services stay up together. Launch with `~/bin/tricorder start (sshd on 8022)`.

9. Install and first run

Uninstall the previous Mk I package before sideloading 0.6.17 (same applicationId). Grant mic, camera, location, nearby as prompted. Install Termux from F-Droid, not Play. On first Earth-link boot the APK fetches BirdNET, flora packs, a translator pair, TLEs, SOL weather, and the dictation embedder file.

Chat/vision/Imagine/Kokoro still need the Termux script and the GGUF/ONNX weights already documented in earlier revisions.

10. Troubleshooting additions for 0.6.15–0.6.17

Symptom	Likely cause	Action
Said dictation; Gemma answered	Build 0.6.15, or a long sentence containing the word	Install 0.6.16+. Say only “dictation”.
No directory after wake	Not in Inquiry, or mic error	Press I, mic listening, speak again.
EDIT seven misses	Recognizer wrote “edit 7” vs “edit to”	Say the digits, or “edit 0007”.
CURSOR MINUS no hit	Keyword not in the buffer, or extra words after the command	Say only CURSOR MINUS <word>.
BACKSPACE ate the wrong word	Cursor was not after the intended token	MINUS/PLUS first, watch , then BACKSPACE.
CUT then PASTE empty	Cursor was already at end of sentence	Place before the sentence, then CUT.
MINDMAP empty	No memos yet, or no overlapping synset	Create two notes that share a cluster word.
SUMMARIZE is extractive only	Chat :8080 down	<code>~/bin/tricorder status</code> ; start chat.
Memos gone after reinstall	Private app storage wiped	Expected. Copy memos.json off-device if you need them.
All backend lamps red	Loopback probes failing	<code>~/bin/tricorder status</code> ; start. Do not kill 8188.

11. Competing products (unchanged conclusion)

Phone “AI” products in 2026 are almost all cloud chat with optional on-device STT/TTS. PocketPal and similar apps run one local GGUF or one TTS graph. Note-taker apps run Whisper then a summarizer. None combine concurrent local chat, vision, turbo diffusion, Kokoro, YAMNet/BirdNET/flora, offline translation, sentry/audio watch, mesh alerts, and a wake-word token-buffer editor on a Snapdragon-class SoC with no cloud in the inference path. That is still the gap this instrument occupies.

12. Revision history

Rev	Date	Notes
0.5.4	4 Sep 2026	Translator, horizon, NOAA/OpenSky, FAA warning
0.6.4	5 Sep 2026	G-meter, GPS speed, Sat/Stars, civic watch, first-run fetch
0.6.8	6 Sep 2026	SOL SYSTEM three-step cycle, SWPC + HamQSL + NWS
0.6.14	8 Sep 2026	Sentry weapon watch, IPL + Kokoro kill, audio soundscape watch
0.6.17	9 Sep 2026	Dictation wake word, memo directory, ZULU, semantic mindmap, token-buffer cursor editor
0.6.18	9 Sep 2026	LOG nnnn converts a dictation memo into a field-log entry after CONFIRMED
0.6.21	10 Sep 2026	Punctuation, live horizon IMU, baro altitude, aircraft units + city pair, ROBO, DIAGNOSTICS

End of Rev 0.6.21. Keep this file with the APK and the project zip. The on-device Help link opens a short briefing; this document is the full operations manual.

4.10 LOG nnnn (0.6.18)

From the dictation directory, say LOG and a memo number (spoken numbers work). The instrument asks: convert nnnn %title% to tricorder log entry? say confirmed or abort. CONFIRMED writes one field-log row, kind dictation, text “DICTATION nnnn title” plus the body, with the same UTC/local timestamp and GPS fix the rest of the log uses. ABORT returns to the directory. The original memo remains. Field-log text is capped at 2 000 characters; very long notes should be summarized first. The CRT footer and the ten-second LED prompt now include LOG beside NEW and EDIT.

4.11 Punctuation in EDIT (0.6.19)

Android STT spells the word “period.” In dictation EDIT say PUNCTUATION and then PERIOD, DOT, COMMA, SEMICOLON, QUESTION (or QUESTION MARK), COLON, or EXCLAMATION. The mark is glued to the token before the cursor: “the rover rolled” + PUNCTUATION PERIOD becomes “the rover rolled.” Isolated PERIOD, DOT, COMMA, SEMICOLON, and QUESTION also convert so the operator is not forced to use the prefix. NEWLINE and BACKSPACE are unchanged.

8.3 Artificial horizon live feed and barometric altitude (0.6.20)

Earlier builds drew the horizon from WebView DeviceMotion. On the Fold 3 that stream often stayed at a default attitude until the operator toggled modes. 0.6.20 registers the platform gravity, accelerometer, and rotation-vector sensors in the host Activity and pushes gx/gy/gz and heading into the canvas about twenty times a second from cold start. The Fold 3 has a barometer, not a certified altimeter. The tape prints BARO in hectopascals and a standard-atmosphere pressure altitude ($44330 \times (1 - (P/1013.25)^{0.1903})$) labeled ALT, plus GPS altitude when the fix includes it. The display remains an uncertified field curiosity. It is not an FAA or NASA attitude instrument and must never be used to fly, land, or station-keep any vehicle.

8.4 Aircraft speech, units, and city pair (0.6.20–0.6.21)

OpenSky's live state vector gives callsign, barometric altitude in meters, and speed in meters per second. It does not include a city pair. 0.6.20 rewrites that clause before robot speech: default metric is "tail ID DAL123 3456 meters 576 kilometers per hour." Settings → Aircraft / altitude English units stores `tricorder.units=english` and yields feet and miles per hour. Aircraft alerts use the cheap robot voice only; Kokoro is not used for overhead traffic. 0.6.21 looks up each new tail: first `adsbdb` callsign (city + IATA when published), then OpenSky flights/aircraft for the last sixteen hours (ICAO departure/arrival). The spoken tail is then "... from Atlanta ATL to Dallas DFW." Routes are cached per hex/callsign. General aviation with no filed route is announced with altitude and speed only. Lookups need earth link. Do not treat the pair as ATC clearance.

8.5 IPL row, ROBO, DIAGNOSTICS (0.6.20)

The IPL sliders are shortened and labeled ALL (abort chat, vision, imagine, voice, translate) and STT (kill in-flight Kokoro only). Beside them is ROBO, a green toggle. When ROBO is on, Inquiry and other `speak()` paths use the system robot voice instead of Kokoro so the operator can run the stack at alert speed. The preference persists in local storage. Saying DIAGNOSTICS (that word) in Inquiry does not go to Gemma. The robot reads software version, device model, battery, earth-link state, barometer, compass heading, accelerometer/gyro presence, Chat/Vision/Image/Voice lamps, BirdNET / flora / translator / Termux pack state, the last init-fetch notes, a short capability and model list, the N2NHU Labs credit, and this disclaimer: Project Tricorder is not authorized by or affiliated with The Wand Company, Paramount SkyDance, Desilu, Gulf and Western, CBS, Roddenberry Entertainment, or Wah Chang, which are trademarks of their respective holders.